18 The straight line $\mathbf{L}_1$ passes through the points with coordinates (4, 6) and (12, 2)
The straight line $\mathbf{L}_2$ passes through the origin and has gradient -3

The lines $\mathbf{L}_1$ and $\mathbf{L}_2$ intersect at point P.

Find the coordinates of P.

Gradient of $L_1 = \dfrac{2 - 6}{12 - 4} = -0.5$

1 mark
Find the gradient

L_1: $y = -0.5x + c$

Using (4, 6): $6 = -2 + c$, so $c = 8$

L_1: $y = -0.5x + 8$; L_2: $y = -3x$

1 mark
Find the equation of L_1

At the intersection, both y-values are equal:

$$-0.5x + 8 = -3x$$
$$(+3x) \qquad (+3x)$$
$$2.5x + 8 = 0$$
$$(-8) \qquad (-8)$$
$$2.5x = -8$$
$$(\div 2.5) \qquad (\div 2.5)$$

1 mark
Equate the two lines and solve

$x = -3.2$; $y = -3(-3.2) = 9.6$

(..... −3.2 ..... , ... 9.6 .....)

1 mark
Both coordinates correct

(Total for Question 18 is 4 marks)